Highly Available 3-Tier Web Application on AWS

Project Overview

This project demonstrates the deployment of a **highly available, scalable, and secure 3-tier web application** on AWS.

The architecture is designed to:

- Handle traffic spikes (auto scaling)
- Ensure high availability (multi-AZ)
- Provide secure access (HTTPS)
- Separate concerns using 3-tier design

Objectives

- Deploy infrastructure across **multiple Availability Zones**
- Configure **Auto Scaling Group (ASG)**
- Use **Application Load Balancer (ALB)** for traffic distribution
- Store static content in S3
- Secure application using HTTPS
- Implement secure access using Bastion Host
- Enable EBS encryption using KMS

Architecture Diagram (Explain in words)

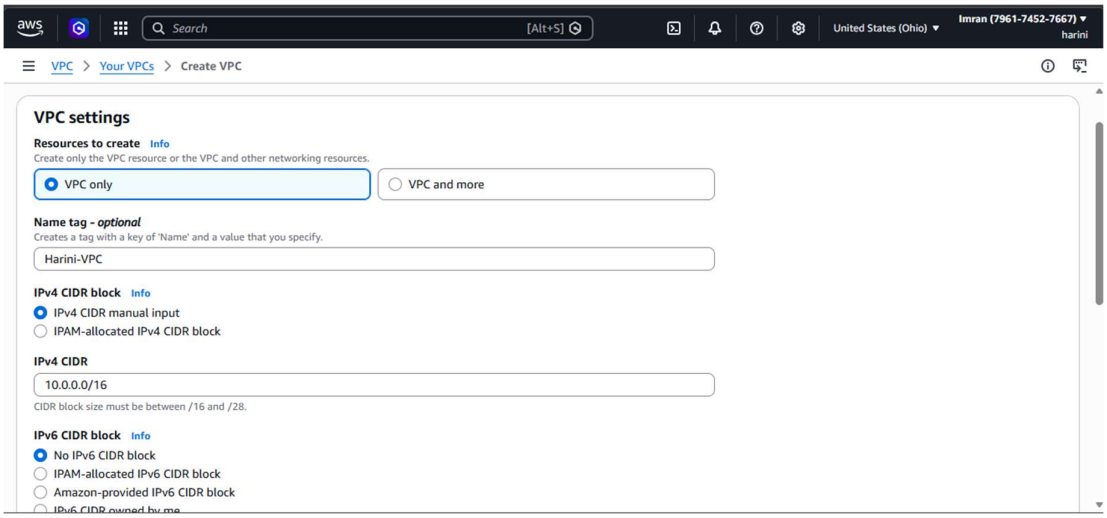
Flow:

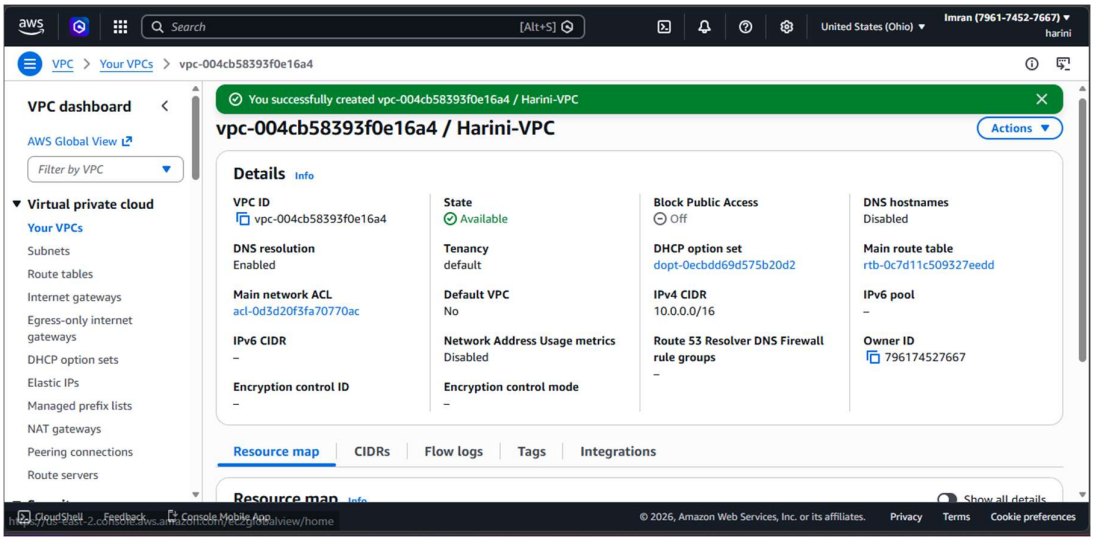
User → ALB (HTTPS) → Target Group → EC2 (ASG in Private Subnets)
↓
EBS
↓
S3
→ Access Private EC2

Infrastructure Setup

VPC Configuration

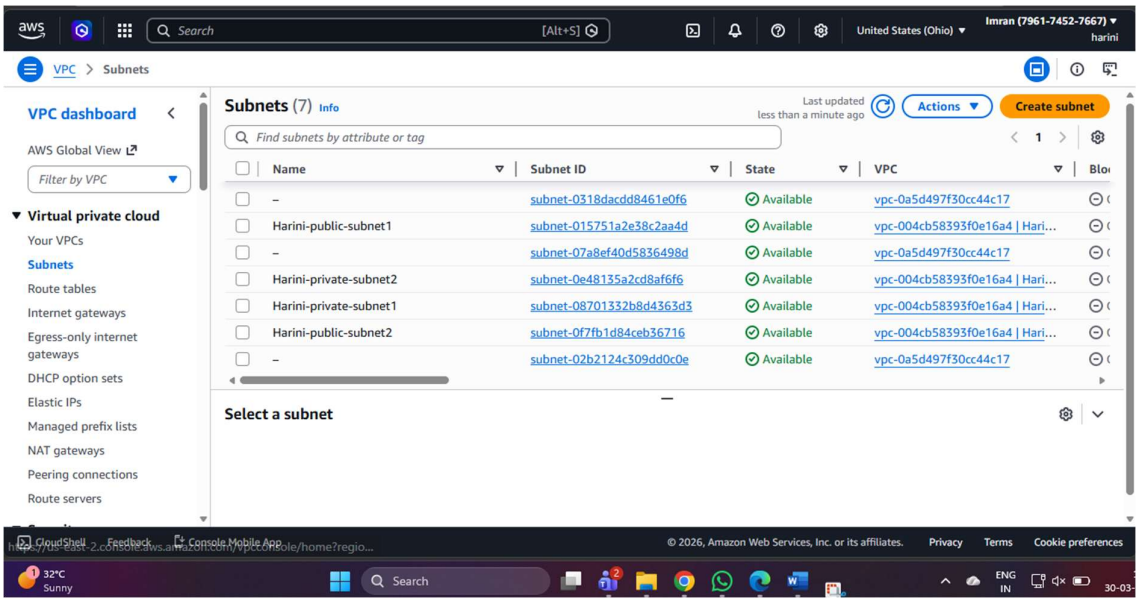
- Created VPC with CIDR: 10.0.0.0/16





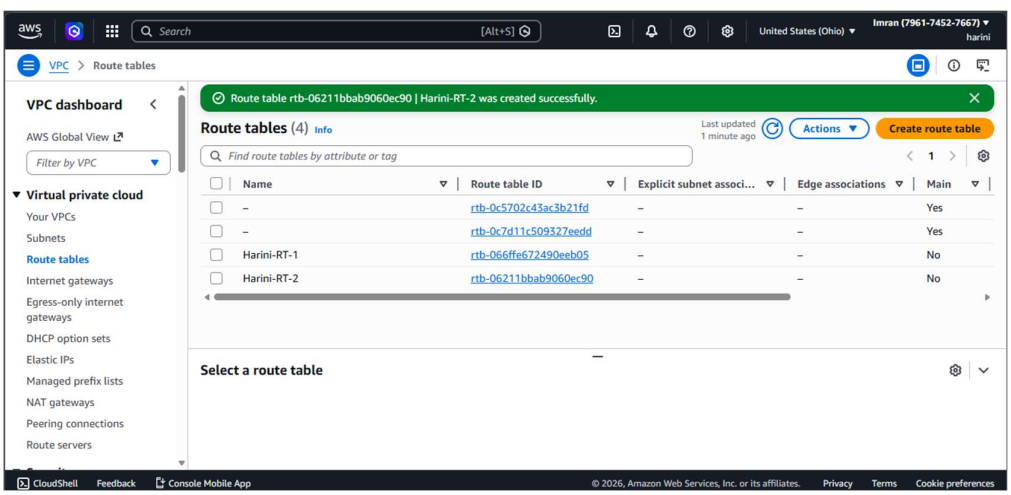
Subnets

- Public Subnets:
  - 10.0.1.0/24 (AZ-1a)
  - 10.0.2.0/24 (AZ-1b)
- Private Subnets:
  - 10.0.3.0/24 (AZ-1a)
  - 10.0.4.0/24 (AZ-1b)



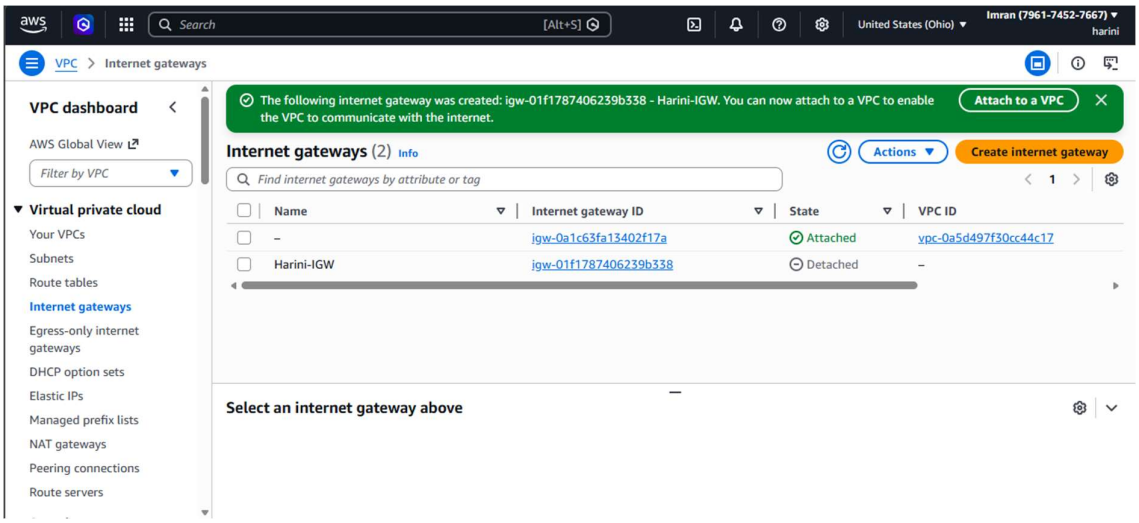
Route Tables

- **Public Route Table (Harini-RT-1)**
  - Route: 0.0.0.0/0 → Internet Gateway
  - Associated with Public Subnets
- **Private Route Table (Harini-RT-2)**
  - Route: 0.0.0.0/0 → NAT Gateway
  - Associated with Private Subnets



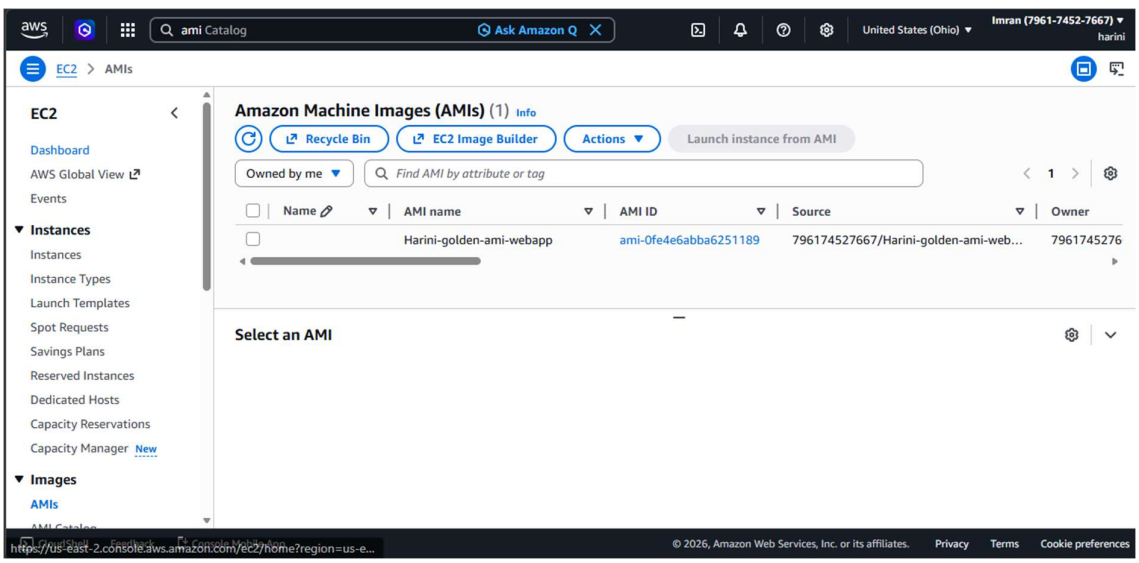
Internet & NAT Setup

- Internet Gateway attached to VPC
- NAT Gateway created in Public Subnet



Golden AMI

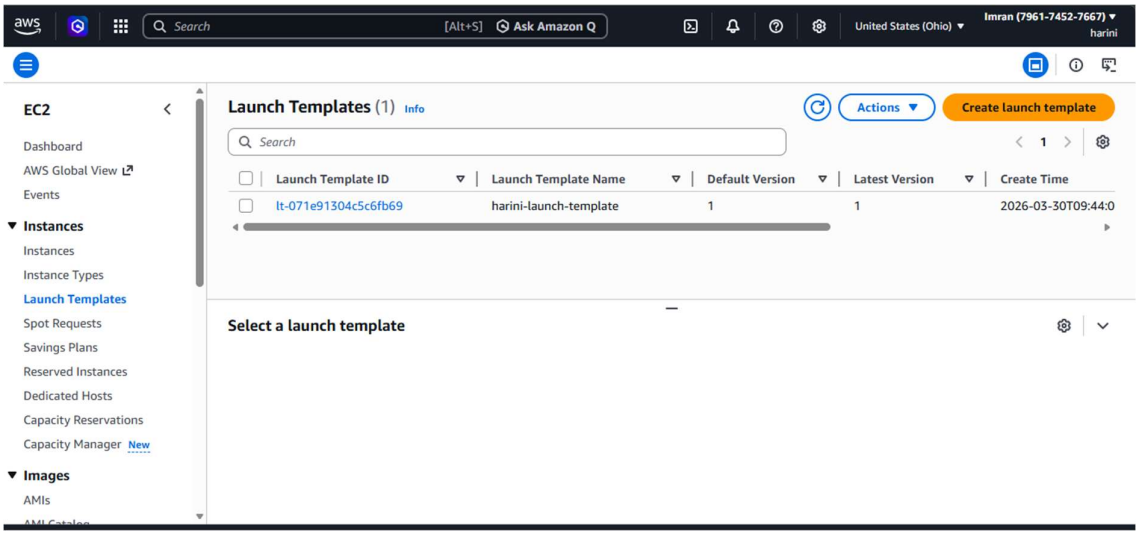
- Installed Apache/Nginx
- Deployed application code
- Created reusable AMI for consistency



Launch Template

- Instance Type: t3.micro
- AMI: Golden AMI
- Security Group attached

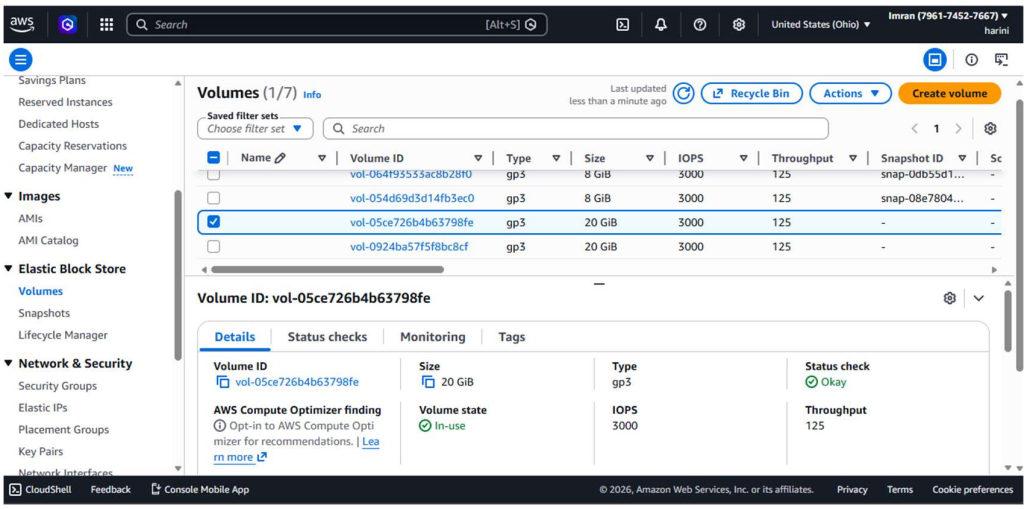
- Configured User Data for automation



Storage Layer

EBS Configuration

- Volume Type: gp3
- Size: 20 GB
- Encryption enabled using AWS Key Management Service



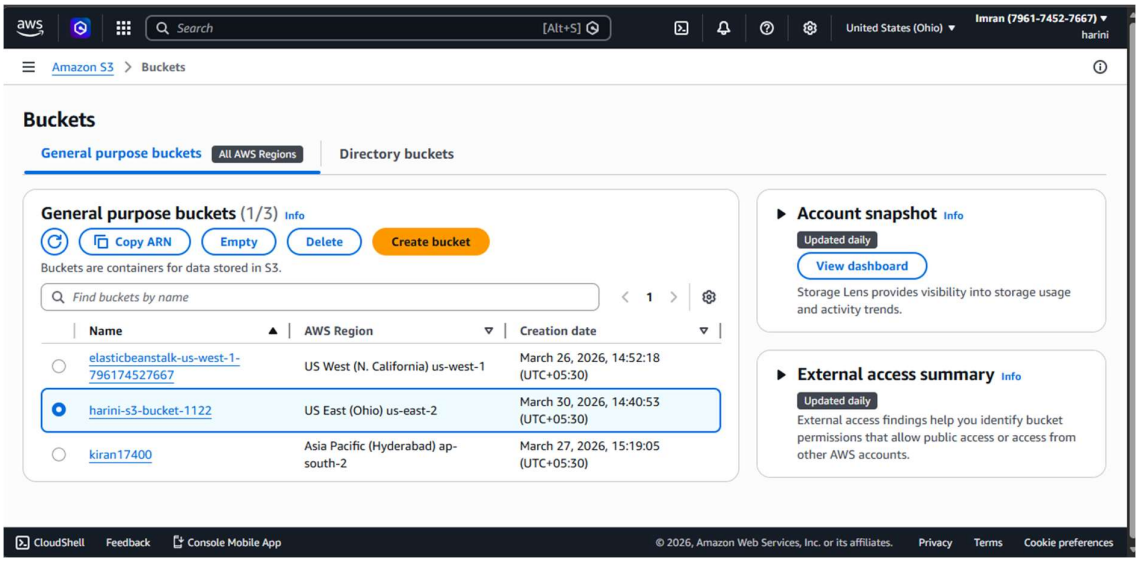
S3 Configuration

Used Amazon S3 for:

- Static files (images, CSS, JS)

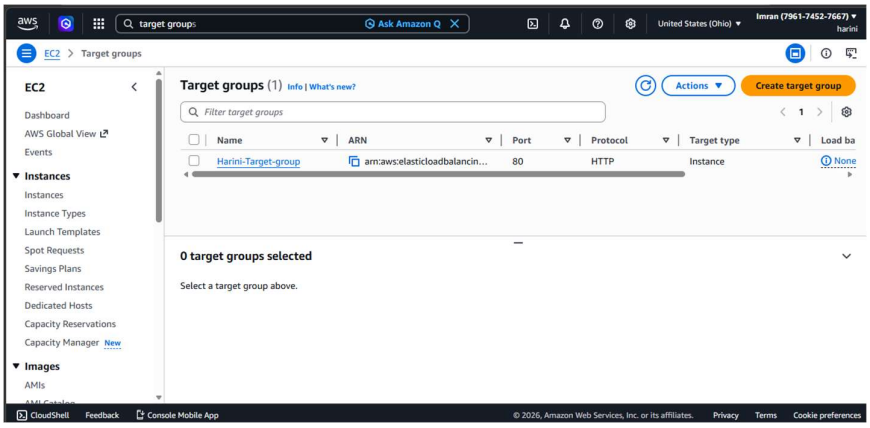
Features:

- Versioning enabled
- Highly durable storage



Target Group

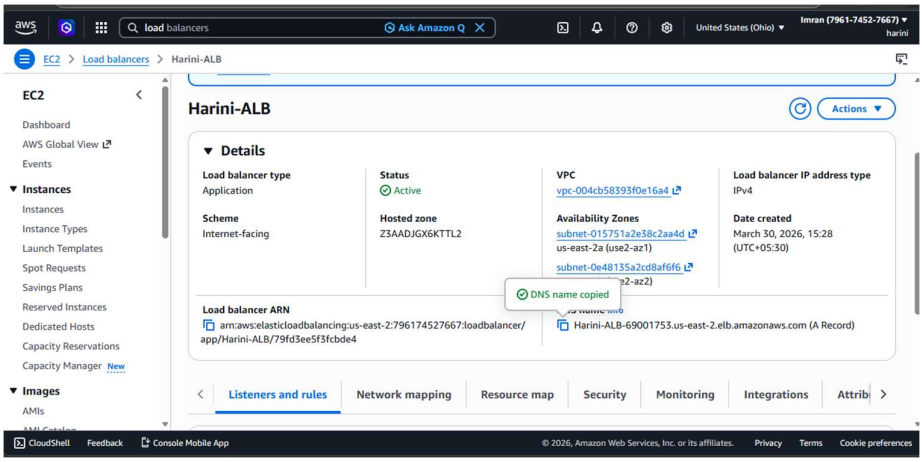
- Protocol: HTTP
- Port: 80
- Health checks enabled



Load Balancer Layer

Application Load Balancer (ALB)

- Type: Internet-facing
- Deployed in Public Subnets



Auto Scaling Group (ASG)

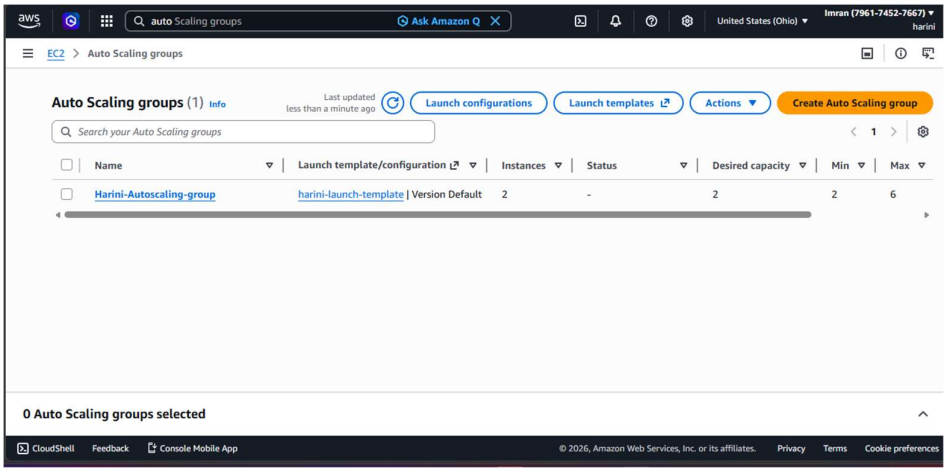
- Min: 2 | Desired: 2 | Max: 4
- Deployed in Private Subnets across 2 AZs

Scaling Policy:

- Target tracking based on CPU utilization (70%)

Features:

- Self-healing (replaces unhealthy instances)
- High availability



HTTPS Configuration

Used AWS Certificate Manager

- Listener: HTTPS (443)

- SSL certificate attached

Advanced Features

- Cross-Zone Load Balancing enabled
- Sticky Sessions enabled (1 hour)

Security Configuration

Security Groups

ALB SG

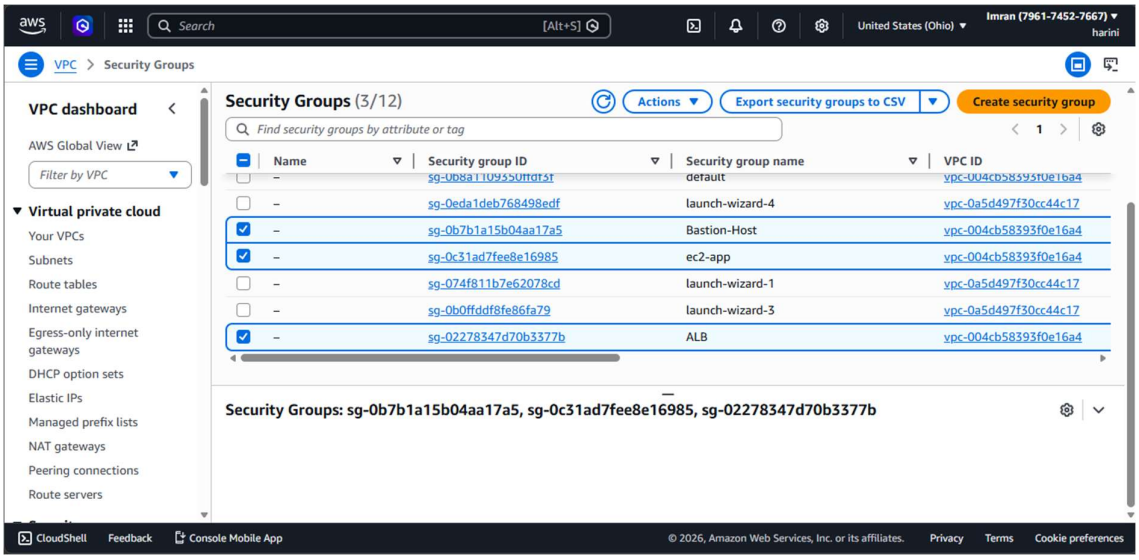
- Allow HTTPS (443) from Internet

App Server SG

- Allow HTTP only from ALB SG

Bastion SG

- Allow SSH (22) only from My IP

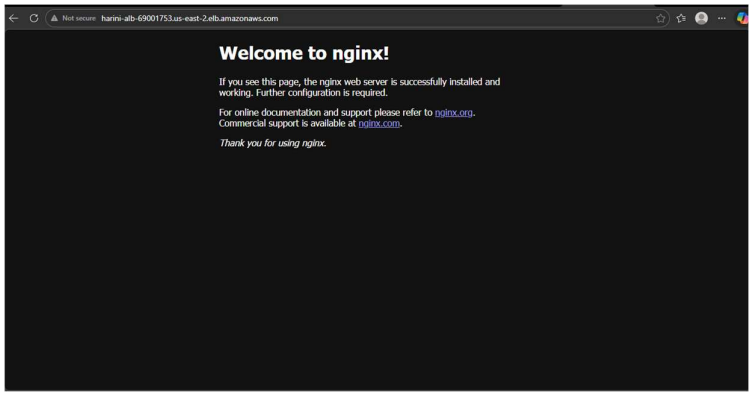


Access Flow

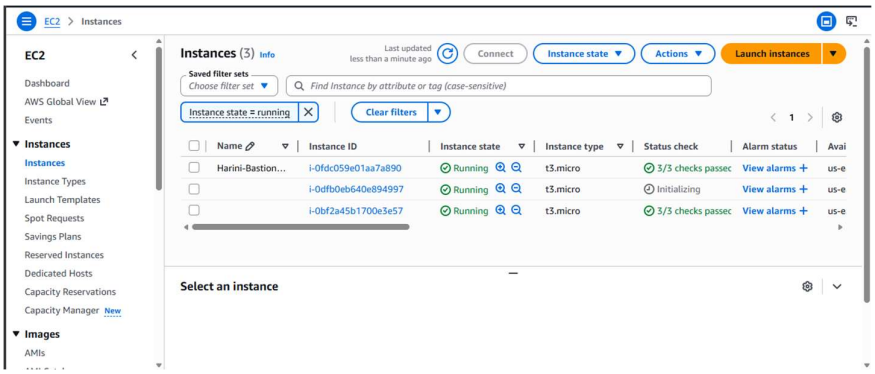
User → ALB → EC2 (Private)
Admin → Bastion → EC2

Validation & Testing

- Accessed application via ALB DNS



- Verified load balancing across instances
- Tested auto scaling using load
- Terminated instance → ASG recreated it
- Verified HTTPS access



Key Concepts Used

- High Availability (Multi-AZ)
- Auto Scaling
- Load Balancing
- Secure Access (Bastion Host)
- Data Encryption (EBS + KMS)
- Object Storage (S3)
- HTTPS Security

Challenges Faced

(You can customize this section)

Example:

- Incorrect route table association
- Security group misconfiguration
- Target group health check failures

Conclusion

The project successfully demonstrates a **scalable, secure, and highly available AWS architecture** capable of handling production-level workloads.

It follows best practices like:

- Multi-AZ deployment
- Auto scaling
- Secure networking
- Encrypted storage